

Edison Ricardo Altamirano Avila

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TECHNICAL SKILLS

AMS/RF/IC: analog IC design, mixed-signal circuits, LDOs, bandgap references, OTAs, gm/Id sizing, PLLs, RF receivers, I/Q demodulation, UWB radar
EDA/Verification: Cadence Virtuoso, layout, DRC/LVS, timing closure, RTL-to-GDS, SKY130, TSMC 180 nm, MATLAB sweeps, simulation/debug
Embedded/Sensing: C/C++, Python, MATLAB, Simulink, RP2040, PX4, IMUs, sensor modeling, signal processing, Git

EXPERIENCE

- Navigation Simulation Engineer** June 2026 – Present
ANELLO Photonics *California, United States*
– Built a navigation simulation framework for inertial sensors and navigation algorithms across fixed-wing aircraft, ground rovers, unmanned surface vehicles, and unmanned underwater vehicles.
– Implemented realistic IMU behavior, sensor models, vehicle dynamics, control integration, data interfaces, validation tools, and reusable simulation components.
– Developed Software-in-the-Loop and Hardware-in-the-Loop workflows using MATLAB, Simulink, PX4, Python, C/C++, and Git for repeatable pre-deployment testing.
- Application Engineer** Nov. 2023 – Sept. 2025
ZF Group *Monterrey, Mexico*
– Contributed to next-generation commercial-vehicle braking systems through model-based development, Software-in-the-Loop validation, system architecture analysis, and embedded software verification.
– Connected requirements, software architecture, Simulink models, component interfaces, signal flow, control logic, and verification results within the automotive V-model process.
– Supported debugging and validation of safety-critical embedded behavior using MATLAB/Simulink, embedded C/C++, AUTOSAR concepts, and automotive development tools.
- Autonomous Vehicle Intern** Aug. 2022 – June 2023
ZF Group *Monterrey, Mexico*
– Integrated CAN bus control, vehicle actuation, visual SLAM, localization, sensor fusion, perception, and autonomy software for an autonomous passenger shuttle.
– Bridged embedded vehicle hardware, robotics algorithms, and real-vehicle testing for a multidisciplinary autonomous systems platform.
- Undergraduate Research Assistant** Jan. 2020 – July 2022
Tecnologico de Monterrey *Monterrey, Mexico*
– Developed robotics and Industry 4.0 research systems involving AGV robots, PLC integration, digital twins, simulation, VR interfaces, and cyber-physical factories.
- Robotics Developer** Aug. 2019 – June 2023
VantTec / RoBorregos *Monterrey, Mexico*
– Developed autonomy software for marine, humanoid, and service robots: ROS, Python, C++, computer vision, SLAM, navigation, state machines, task planning, and sensor integration.

SELECTED PROJECTS

- EE372 Analog/Mixed-Signal LDO Chip** | TSMC 180 nm, Cadence, bandgap, OTA, I2C, DRC/LVS
– Designed and laid out a mixed-signal LDO with BANBA bandgap reference, 5-bit trim network, high-gain error amplifier, Schmitt-trigger protection, CMOS MUX, I2C trim control, full-chip integration, and incremental DRC/LVS closure; earned 10/10 and presented at Apple.
- EE233 FM Superheterodyne Receiver** | RF, PLL, I/Q demodulation, embedded control
– Built a dual-conversion FM receiver with Si5351 PLL synthesis, quadrature demodulation, RP2040 firmware, analog signal-chain debugging, and live browser tuning interface.
- EE219 UWB Radar Processing** | radar, CFAR, Kalman/EKF, vital signs
– Built a pulsed-UWB radar pipeline for CIR calibration, clutter removal, OS-CFAR, Doppler processing, Kalman/EKF tracking, Hungarian assignment, people counting, and breathing-rate extraction.

EDUCATION

- Stanford University** 2024 – Present
M.S. Electrical Engineering; Arbabian Lab from start of M.S.; RF/analog, mixed-signal IC, VLSI, radar sensing *Stanford, CA*
- Tecnologico de Monterrey** 2018 – 2023
B.S. Digital Systems and Robotics; robotics, mechatronics, automation, embedded systems, and research *Monterrey, Mexico*

SELECTED PUBLICATIONS AND AWARDS

IEEE/Springer publications in digital twins, robot operating systems, mixed/virtual-reality industrial robotics, and mobile-manipulator navigation. Awards: AIRLab Stacking Challenge Most Consistent Solution; IROS 2022 HandyMan Challenge 3rd place; RoboBoat 1st place and RoboSub 3rd place.